

Mathematics for Machine Learning

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Elementary transformations

Definition

Given a SLE

[illegible]

$$A := \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

is called the **coefficient matrix** and

$$\left[\begin{array}{c|c} A & \mathbf{b} \end{array} \right]$$
 is called the **augmented matrix**,
$$\text{where } \mathbf{b} = [b_1, b_2, \dots, b_m]^T$$

Definition

A matrix is in **row echelon form (REF)** if it satisfies the following properties:

1. Any rows consisting entirely of zeros are at the bottom.
2. In each nonzero row, the first nonzero entry (called the **leading entry** or **pivot**) is in a column to the left of any leading entries below it.

Elementary Row Operations

Example

The following matrices are in row echelon form:

$$\begin{bmatrix} 1 & 3 & 5 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & -2 & 7 \\ 0 & 4 & 1 \\ 0 & 0 & 3 \end{bmatrix} \quad \begin{bmatrix} 3 & 1 & -2 & 7 \\ 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 4 \end{bmatrix}.$$

Assuming that each of these matrices is an augmented matrix, write out the corresponding SLE and solve them.

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Assuming that each of these matrices is an augmented matrix, write out the corresponding SLE and solve them.

Definition

*The following **elementary row operations** can be performed on a matrix:*

- 1. Interchange two rows.*
- 2. Multiply a row by a nonzero constant.*
- 3. Add a multiple of a row to another row.*

Elementary Row Operations

Shorthand notation

1. $R_i \leftrightarrow R_j$ means interchange rows i and j .
2. kR_i means multiply row i by k .
3. $R_i + kR_j$ means add k times row j to row i (and replace row i with the result).

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Example

Reduce the following matrix to echelon form:

$$\begin{bmatrix} 1 & -1 & 2 & 4 \\ 3 & -1 & 8 & 15 \\ -2 & 0 & -4 & -2 \end{bmatrix}.$$

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*The variables corresponding to the pivots in the row-echelon form are called **basic variables** or **leading variables**, the other variables are **free variables**.*

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Example

Let the REF of the augmented matrix be

$$\begin{bmatrix} 1 & -1 & 2 & 4 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The leading variables in row echelon form are x_1, x_3 , and the free variable is x_2 .

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Example

Solve the system

$$\begin{cases} 2x_1 - x_2 + 5x_3 = -2 \\ x_1 - 2x_2 + 4x_3 = -7 \\ 3x_2 - 2x_3 = 9 \end{cases}.$$

Definition

A matrix is in **reduced row echelon form** if it satisfies the following properties:

1. It is in row echelon form.
2. The leading entry in each non-zero row is a 1 (called a **leading 1**).
3. Each column containing a leading 1 has zeros everywhere else.

The entire process is called **Gauss-Jordan elimination**.

Gauss-Jordan Elimination

Example

The following matrix is in reduced row echelon form:

$$\begin{bmatrix} 0 & 1 & 3 & 0 & 0 & -1 & 4 & 0 & 6 \\ 0 & 0 & 0 & 1 & 0 & 3 & -2 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 2 & 2 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Gauss-Jordan Elimination

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Example

Solve the system by Gauss-Jordan elimination.

$$\begin{cases} w - 2x + 3y + 2z = 1 \\ 2w - 4x + 7y + 2z = 4 \\ -3w + 6x - 7y - 10z = 1 \end{cases}.$$

The Gauss-Jordan Method for Computing the Inverse

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Fundamental Theorem of Invertible Matrices

Let A be an $n \times n$ matrix. The following statements are equivalent:

- 1 A is invertible.
- 2 $A\mathbf{x} = \mathbf{b}$ has a unique solution for every vector $\mathbf{b} \in \mathbb{R}^n$.
- 3 $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- 4 The reduced row echelon form of A is I_n .
- 5 A is a product of elementary matrices.

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Theorem

Let A be a square matrix. If a sequence of elementary row operations reduces A to I , then the same sequence of elementary row operations transforms I into A^{-1} .

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If A is row equivalent to I , then elementary row operations will yield

$$[A \mid I] \rightarrow [I \mid A^{-1}].$$

If A cannot be reduced to I , then the Fundamental Theorem guarantees us that A is not invertible.

Examples

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Find the inverse of

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 5 \\ -3 & 2 & 2 \end{bmatrix}$$

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Remark

It can be proved that $\det(A) = \sum_{i=1}^n (-1)^{i+j} a_{ji} \det(A_{ji})$.

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Compute the following determinant

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}.$$

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Example

Compute determinant of an upper triangular matrix.

Invertible Matrix Theorem

Theorem

For any square matrix $A \in \mathbb{R}^{n \times n}$ it holds that A is invertible if and only if $\det(A) \neq 0$.

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- If $A = QBQ^{-1}$ then $\det(A) = \det(B)$.

Vector Spaces and Subspaces

Vector Spaces and Subspaces

Definition

Let V be a set on which addition and scalar multiplication have been defined. If the following axioms hold for all $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ in V and for all scalars c and $d \in \mathbb{R}$ then V is called a **vector space** and its elements are called **vectors**.

1. $\mathbf{u} + \mathbf{v}$ is in V .

Closure under addition

2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$

Commutativity

3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$

Associativity

4. $\exists \mathbf{0} \in V$, (called a **zero vector**), s.t. $\mathbf{u} + \mathbf{0} = \mathbf{u}$

5. $\forall \mathbf{u} \in V, \exists -\mathbf{u} \in V$

s.t. $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$

6. $c\mathbf{u} \in V$

Closure under scalar mult.

7. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$

Distributivity

8. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$

Distributivity

9. $c(d\mathbf{u}) = (cd)\mathbf{u}$

10. $1\mathbf{u} = \mathbf{u}$

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For any natural m and n , the set of all $m \times n$ matrices $\mathbb{R}^{m \times n}$ forms a vector space with the usual operations of matrix addition and matrix scalar multiplication. Here the "vectors" are actually matrices.

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Example

*The set $\mathbb{Z}$ of integers with the usual operations is **not** a vector space.*

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$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3, \quad q(x) = b_0 + b_1x + b_2x^2 + b_3x^3$$

are in $\mathcal{P}_3$, then

$$\begin{aligned} p(x) + q(x) &= \\ &= (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + (a_3 + b_3)x^3 \end{aligned}$$

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In general, for any fixed $n \geq 0$, the set $\mathcal{P}_n$ of all polynomials of degree less than or equal to n is a vector space, as is the set $\mathcal{P}$ of all polynomials.

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A line through the origin is a subspace of $\mathbb{R}^2$.

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Theorem

Let V be a vector space and let W be a nonempty subset of V . Then W is a subspace of V if and only if the following conditions hold:

- a. If $\mathbf{u}$ and $\mathbf{v}$ are in W , then $\mathbf{u} + \mathbf{v}$ is in W .*
- b. If $\mathbf{u}$ is in W and c is a scalar, then $c\mathbf{u}$ is in W .*

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- The intersection of arbitrarily many subspaces is a subspace itself.

Example

Which of the following is a subspace of $\mathbb{R}^2$?

- 1 *a line passing through the origin*
- 2 *two distinct lines passing through the origin*
- 3 *a unit circle*
- 4 *a line not passing through the origin*